INTERNAL TECHNICAL DOCUMENT — PRIMORDIAL ALGORITHM PROJECT

Time, Energy, Superposition & Collapse

A unified perceptual-physical account

On what time actually is,
why energy is its avatar,
why superposition is temporal not spatial,
and what collapse threshold really means

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These ideas were developed through active reasoning sessions,
not derived from existing literature.
Cross-references to physics are confirmations, not sources.

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1. Preamble: Method

The ideas in this paper were not derived from textbooks. They were derived from one principle:

Core method

Every mechanism of motion and transformation repeats across systems. Only the meaning changes per domain.
To understand a phenomenon: find the abstract pattern, map it to the domain, verify the mapping holds mechanically. If it does, the pattern is real.

This paper collects four such mappings developed in sequence: time as material state change, energy as avatar of time, superposition as temporal overlap, and collapse as field displacement threshold. Each section builds on the previous.

2. Time Is Not a Thing

2.1. The standard picture and its problem

Physics treats time as a coordinate — a dimension t alongside x, y, z. But this creates a question that physics does not answer: *what is time made of?* Unlike x, y, z which describe positions of matter, t has no material referent.

The answer from perception: time is not a thing. It is a **label that the brain assigns to the detected difference between two material states.**

2.2. Formal statement

Definition 2.1 (Material state)**.** *A **material state** at moment τ is the full configuration of a physical system:*

$$\mathcal{S}(\tau) = \{(m_i,\, \mathbf{x}_i,\, \mathbf{p}_i)\}_{i=1}^{N}$$

the set of masses, positions, and momenta of all N constituents.

Definition 2.2 (Time as state differential)**.** ***Time** is not a dimension. It is the perceived label for the difference between two material states:*

$$\Delta t \;\propto\; \|\mathcal{S}(\tau_1) - \mathcal{S}(\tau_0)\|$$

No material change $\Rightarrow$ no time. Time is the measure of $\mathcal{S} \to \mathcal{S}'$.

Consequence

A clock does not measure time. A clock measures **cycles of material state change** — oscillation of quartz, decay of caesium atoms, pendulum swing. We call the count of those cycles "time passed." The label is convenient. The underlying thing is always material change.

2.3. Perceptual rendering: discrete to continuous

Consider a ball being deformed. At the physical level:

- Matter (molecules, atoms) changes position in 3D space
- The change is discrete: finite number of particles, finite position shifts, finite energy transfer
- There is no "continuous motion" at the physical level

At the perceptual level:

- The eye receives photons reflected from the ball at t_0 and t_1
- The brain computes the **diff** between two photon patterns: $\Delta(\text{photon}_{t_0}, \text{photon}_{t_1})$
- Because the sampling rate of the visual system (≈24 Hz conscious, higher unconscious) exceeds the rate of large-scale material change, the gaps between discrete samples are **interpolated** into smooth motion

The brain does not see motion. The brain sees two states and renders the gap between them as motion.
This is why cinema works at 24 fps. This is why a spinning fan blade appears solid. This is why a deforming ball appears to "move smoothly" — the smoothness is a rendering artifact, not a property of the physical process.

2.4. Subjective time dilation

If time = perceived rate of material state change, then subjective time follows directly:

Condition	Material changes	Perceived time
High activity, dense input	Many $\mathcal{S} \to \mathcal{S}'$	Fast
Boredom, low input	Few $\mathcal{S} \to \mathcal{S}'$	Slow
Sleep, no sensory input	Zero detected Δ	Instantaneous
Flow state	Many changes, all attended	Variable

This is not metaphor. This is the mechanism: the brain's time experience is *proportional to the density of state changes it successfully processes and distinguishes.*

3. Energy as Avatar of Time

3.1. What energy actually is

Physics defines energy as a conserved quantity. But what *is* it?

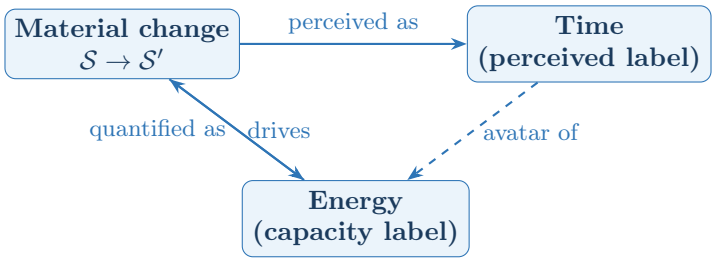
Feynman's honest answer: *"It is important to realize that in physics today, we have no knowledge of what energy is."* Energy is not a substance. It is not a fluid. It is a number that is conserved across state changes.

Definition 3.1 (Energy as state-change capacity)**.** ***Energy*** *is the label assigned to the* ***capacity of a system to cause material state change*** *in itself or in another system.*

$$E \equiv \textit{capacity to produce } \mathcal{S} \to \mathcal{S}'$$

3.2. The circular dependency

If time = perception of $\mathcal{S} \to \mathcal{S}'$, and energy = capacity to cause $\mathcal{S} \to \mathcal{S}'$, then:



Proposition 3.1 (Co-emergence)**.** *Time and energy do not have a causal order. They* ***co-emerge*** *from material state change. Neither creates the other. Both are labels for aspects of*

the same underlying process:

$$\textit{Time} \leftrightarrow \textit{Energy} \leftrightarrow \mathcal{S} \to \mathcal{S}'$$

The question "where does energy come from?" is therefore ill-posed in the same way as "what is north of the North Pole?" — it assumes a prior that does not exist.

3.3. Measurement always goes through material change

This is why energy is always measured indirectly:

- Kinetic energy: measure velocity (position change over time)
- Thermal energy: measure temperature (average particle motion)
- Electrical energy: measure current (charge displacement)
- Nuclear energy: measure mass deficit before/after reaction

In every case: we measure **material change**, then infer energy from the magnitude of that change. Energy is never observed directly. It is always back-calculated from $\Delta\mathcal{S}$.

4. Superposition as Temporal State Overlap

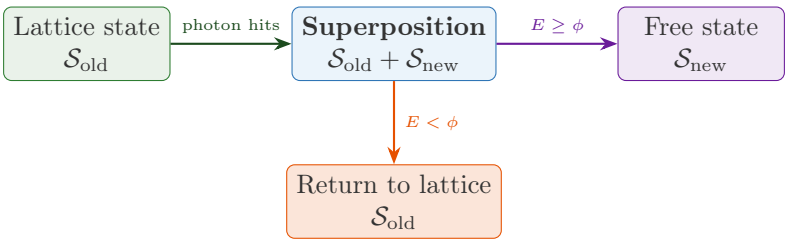
4.1. The standard picture and why it confuses

Quantum superposition is usually described spatially: "a particle is in two places at once." This is technically a consequence of the wave formalism but it is deeply unintuitive and has generated decades of mystical interpretation.

The reframe: superposition is not a spatial paradox. It is a **temporal condition** — a particle carrying two states simultaneously because it has not yet fully committed from one material configuration to another.

4.2. The photoelectric case

In the photoelectric effect (the canonical two-state experiment), an electron bound in a metal lattice is struck by a photon:



At the moment of interaction, the electron is:

- **State 1**: residue of the old system — the energy signature of the lattice it is still bound to, the "memory" of its prior context
- **State 2**: potential of the new system — the energy offered by the photon, the "possible future" of being free

Definition 4.1 (Superposition as temporal overlap)**.** ***Superposition*** *is the condition in which a particle simultaneously carries:*

$$\Psi = \alpha\,|\mathcal{S}_{old}\rangle + \beta\,|\mathcal{S}_{new}\rangle, \qquad |\alpha|^2 + |\beta|^2 = 1$$

where $\mathcal{S}_{old}$ is the residue of prior context (energy signature of the system the particle is leaving) and $\mathcal{S}_{new}$ is the potential of the incoming context (energy signature of the system it may join).

*This is a **temporal transition state**, not a spatial paradox.*

4.3. Double-slit reframed

The double-slit experiment: when a particle is not measured at the slits, it produces an interference pattern. When measured, it does not.

Standard explanation: "observation collapses the wavefunction."

Reframe:

- **Unmeasured**: the particle does not leave a distinguishable $\Delta\mathcal{S}$ in any system at the slit. Both paths remain as temporal potentials. Both $\mathcal{S}_{\text{path A}}$ and $\mathcal{S}_{\text{path B}}$ are carried. Interference occurs.
- **Measured**: the detector creates a distinguishable $\Delta\mathcal{S}$ — the particle's passage alters the detector's material state in a way that encodes which-path information. One temporal state wins. Interference is destroyed.

The key reframe

"Observation" in quantum mechanics means nothing special about consciousness or observers. It means:
An interaction that produces a distinguishable, non-reversible $\Delta\mathcal{S}$ in some other system.
No $\Delta\mathcal{S}$ written $\Rightarrow$ superposition intact.
$\Delta\mathcal{S}$ written $\Rightarrow$ collapse.

5. Collapse as Field Displacement Threshold

5.1. The field model

Consider a system $\mathcal{B}$ consisting of N bound elements $B_1, B_2, \ldots, B_N$ at equilibrium spacing C:

$$\mathcal{B} = \{B_i : \text{spacing} = C,\ i = 1, \ldots, N\}$$

At equilibrium, the elements form a stable interference field — energy propagates through the system at regular intervals, establishing a coherent internal field pattern $\mathcal{F}(\mathcal{B})$.

Now particle A with energy E_A is incident on $\mathcal{B}$.

5.2. Two regimes

5.2.1. Weak perturbation: $E_A < E_{\text{disp}}$

A does not have sufficient energy to displace any B_i from its lattice position. The perturbation propagates into $\mathcal{F}(\mathcal{B})$ but is absorbed and smeared out by the coherent field:

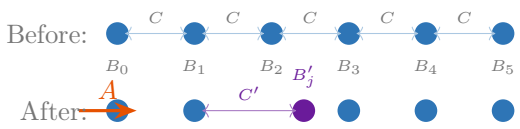
$$E_A < E_{\text{disp}} \;\Rightarrow\; \mathcal{F}(\mathcal{B}) \to \mathcal{F}'(\mathcal{B}) \approx \mathcal{F}(\mathcal{B})$$

The field re-equilibrates to approximately its prior state. No permanent $\Delta\mathcal{S}$ is written. The perturbation is indistinguishable from background fluctuation.

Result: no collapse. Superposition of A is maintained.

5.2.2. Strong perturbation: $E_A \geq E_{\text{disp}}$

A displaces B_j from position x_j to x'_j. The lattice spacing changes locally: $C \to C'$ in the neighborhood of B_j.



With $C \to C'$:

- Energy propagation timing changes: elements now interact at C' instead of C, so collisions occur earlier
- Earlier collision $\Rightarrow$ earlier energy release $\Rightarrow$ local perturbation propagates through field
- The field $\mathcal{F}(\mathcal{B})$ must re-establish at new geometry $\mathcal{F}'(\mathcal{B}')$
- During re-establishment: permanent $\Delta\mathcal{S}$ exists

Definition 5.1 (Collapse threshold)**.** ***Collapse occurs*** *when the incident energy exceeds the displacement energy of the target field:*

$$\boxed{E_A \geq E_{\text{disp}}(\mathcal{B}) \;\Rightarrow\; \textit{collapse}}$$

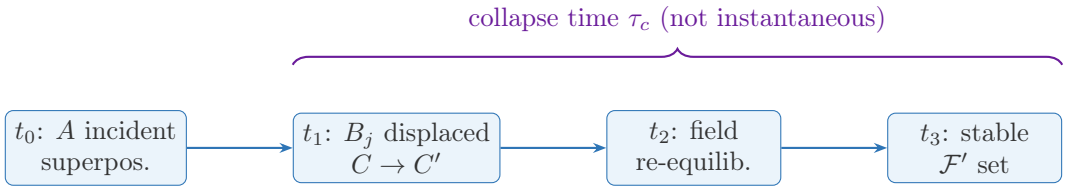
where

$$E_{\text{disp}}(\mathcal{B}) = \textit{minimum energy to shift any } B_i \textit{ such that } C \to C' \textit{ is permanent}$$

Proposition 5.1 (Threshold is relational)**.** *E_{disp} is not a property of A alone or $\mathcal{B}$ alone. It is a property of the* ***relationship*** *between A's energy and $\mathcal{B}$'s internal coherence. The same A may collapse a weak field and leave a strong field unaffected.*

5.3. Collapse is a process, not an event

After displacement (t_1), the field does not instantly settle. It re-equilibrates over a finite timescale:



τ_c is the propagation time of $\Delta\mathcal{S}$ through field $\mathcal{B}$. This is the physical basis of **decoherence timescale** — collapse is not an instantaneous event but a process with a characteristic duration determined by field size and coupling strength.

5.4. Why gravity does not collapse

Corollary 5.1 (Gravitational non-collapse)**.** *For a single particle A, the gravitational perturbation on field $\mathcal{B}$ is:*

$$E_{grav} = \frac{G m_A m_B}{r^2} \cdot \delta r \;\ll\; E_{\text{disp}}(\mathcal{B})$$

for any macroscopic $\mathcal{B}$ at experimentally relevant distances. Therefore gravitational interaction does not collapse superposition, consistent with observation.

This is the same result that Penrose and Diosi reach from objective collapse theory, arrived at here from field displacement dynamics rather than from formal gravitational decoherence.

5.5. Connection to double-slit

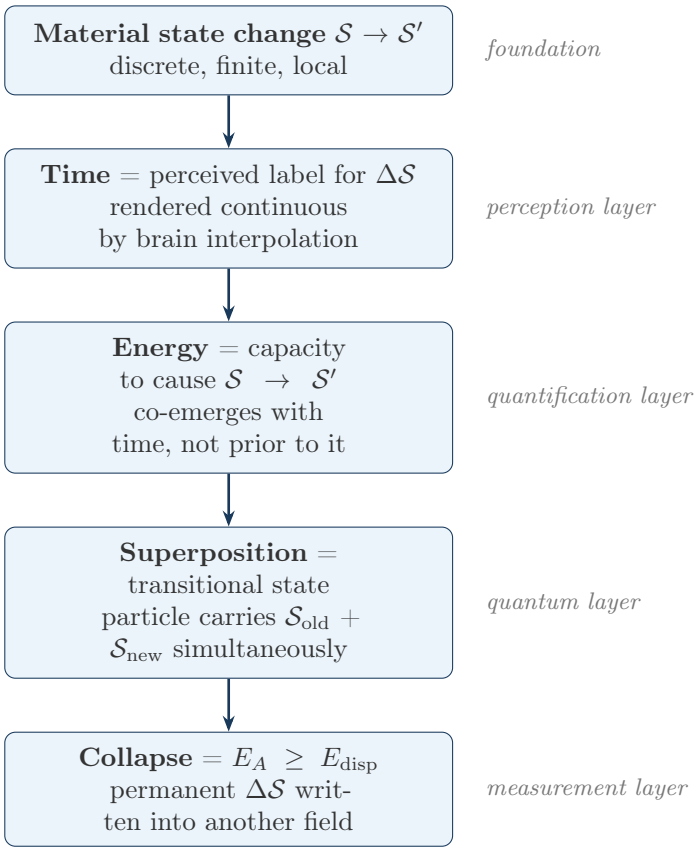
- **No detector at slit**: the particle's passage does not produce $E_A \geq E_{\text{disp}}$ in any system. No $\Delta\mathcal{S}$ permanent enough to be distinguishable. Both paths remain as potentials. Interference pattern emerges on screen.

- **Detector at slit**: the detector's field $\mathcal{B}_{\text{det}}$ is designed so that $E_A \geq E_{\text{disp}}(\mathcal{B}_{\text{det}})$. Particle displaces a field element. $C \to C'$ locally. Which-path information is written into $\mathcal{B}_{\text{det}}$. Interference is destroyed.

"Which type of detector" is irrelevant — photon-based, mechanical, or electric. What matters is only whether $E_A \geq E_{\text{disp}}$ is satisfied.

6. Unified Picture

The four ideas form a single coherent chain:



Single-sentence summary:
Everything that physics calls "time," "energy," "superposition," and "collapse" are four different labels for four different aspects of the same underlying event: **one material configuration transitioning to another.**

7. Open Questions

This paper closes several questions but opens others. Listed here for future development:

1. **What is the initial $\mathcal{S}$?** If time = $\Delta\mathcal{S}$ and energy = capacity for $\Delta\mathcal{S}$, then asking "where did energy come from before the Big Bang" is asking about $\mathcal{S}$ before there was an $\mathcal{S}$. Three possible resolutions: zero-energy universe ($E_{\text{matter}} + E_{\text{gravity}} = 0$),

quantum fluctuation from vacuum, or the question is category-invalid ("north of the North Pole"). No resolution is fully satisfying.

2. **Is E_{disp} derivable from first principles?** Currently defined as a threshold. Can it be calculated from field coherence properties alone, without experimental input?

3. **What is τ_c as a function of field parameters?** Collapse time is real and measurable (decoherence experiments). Can the field displacement model produce a closed-form $\tau_c(N, C, E_A)$?

4. **Consciousness and time perception.** If time is perceived rate of $\Delta\mathcal{S}$, and consciousness is a process that detects and integrates $\Delta\mathcal{S}$, then what is the relationship between the density of conscious processing and the subjective experience of time? Flow states, meditation, and trauma all alter time perception — do they alter $\Delta\mathcal{S}$ detection rate, or the interpolation mechanism, or both?

5. **Primordial integration.** The temporal axis in Primordial v4 is currently implicit (Dan drive accumulation as proxy for time). This paper suggests formalizing it explicitly as accumulated $\Delta\mathcal{S}$ in the V field. V field as substrate = initial $\mathcal{S}$. Dan = process causing $\mathcal{S} \to \mathcal{S}'$. Chuyen = $E_A \geq E_{\text{disp}}$ condition.

8. Map to Primordial Algorithm

This paper	Primordial Algorithm	Note
$\mathcal{S} \to \mathcal{S}'$	Dan drive accumulation	State change = what generates temporal markers
Time = $\Delta\mathcal{S}$	Steps T in simulation	T counts state changes, not clock seconds
V field (substrate)	Initial $\mathcal{S}$ with H1 residue	V is never blank; carries prior cycle memory
Superposition	Chuyen pre-fire state	Entity carries old drive + new potential simultaneously
Collapse	Chuyen fires: $\Delta > \theta$	Threshold = drive_acc > 0.3 in v4
E_{disp}	Chuyen threshold θ_{drive}	Same concept: minimum to create permanent state change
τ_c (collapse time)	Fractal depth propagation	Collapse is not instant; propagates through depth levels
Decoherence	Entity interacting with many others	Too many $\Delta\mathcal{S}$ written; classical behavior emerges

9. Summary

Six claims of this paper

1. **Time is not a dimension.** It is the brain's label for detected material state change. No change ⇒ no time.
2. **Energy is the avatar of time.** Both co-emerge from $\mathcal{S} \to \mathcal{S}'$. Neither is prior to the other. Asking "where did energy come from" is asking for a prior that does not exist.

3. **Physics describes, not explains.** Every physical equation describes a relationship. The primitives (mass, charge, space) are undefined. This is not a gap — it is a necessary feature of any formal system (Godel).
4. **Superposition is temporal, not spatial.** A particle in superposition carries residue of its old state and potential of its new state simultaneously. It is in a transitional condition between two material configurations, not in two places.
5. **Collapse is field displacement threshold.** Collapse occurs when $E_A \geq E_{\text{disp}}(\mathcal{B})$ — when the incident energy is sufficient to permanently alter the internal geometry of the target field. Below threshold: absorbed and smeared. Above: permanent $\Delta\mathcal{S}$ written. "Observation" is just this condition being satisfied.
6. **Collapse is a process with duration τ_c.** Not an instantaneous event. Duration = propagation time of $\Delta\mathcal{S}$ through the field. This is the physical basis of decoherence timescale.

Kevin T.N. · Primordial Algorithm Project · March 2026
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Companion papers: DCF v1.0, ARC & DH Interaction Types